Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate the concern of Back door Sagging & rattle noise in Scorpio W110 Model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**1/8/19**	**1/8/19**
Completion Date	**18/1/20**	**18/1/20**

Current Status: **Closed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

Cross Functional Team

Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Satish Patil	L7-O	STCF	9922448571
2	Mentor	Rahul Narkar	L5-M	STCF	9881080299
3	Member 1	Aher Nitin	L6-M	CAI	9960800220
4	Member 2	Nitin Hire	L7-O	SCORPIO BODY	9881737077
5	Member 3	Dyaneshwar Visapute	L7-O	SCORPIO BODY	7774065744
6	Member 4	Sachin Bhamare	L7-O	MQA	9158960111
7	Member 5	Jayesh Jirekar	L7-0	PROD.	9860961900

Step1. Select Subject

Brief Description :

- What : Back door Sagging & rattle noise at door alignment ,squeak & rattle stage & RFI for W110 Model.
- Where : In CAI during door alignment defect generated in W110 model.
- When : From Aug-19.
- Who : By Door alignment ,Squeak & Rattle stage associate
- Why : Repeat issue in CAI, RFI impacting on customer satisfaction.
- How : Avg 166 incidences per month.

2. Justify the choice (reasons)

PROMISE 2022

AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment

'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV

Doubling volume from overseas markets

Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

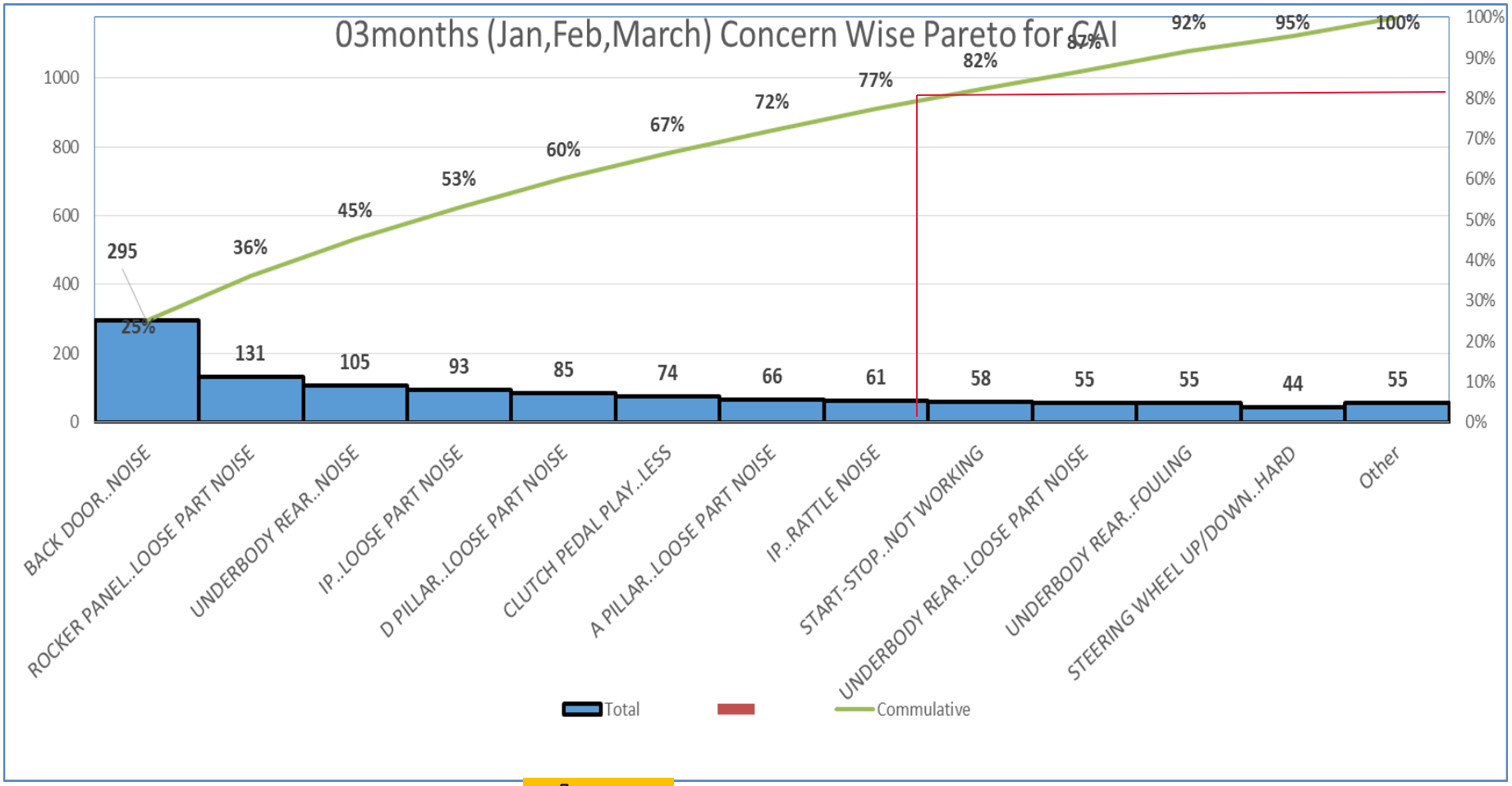
Departmental Objective :-

BIQ – DPR Improvement

- FRC / FBO Improvement

2. Justify the choice

Pareto Analysis for CAI Squeak & Rattle Stage of CAI



Inference – Top defect in Incidences data taken from August -2019

2. Justify the choice

Severity & Consequences of Concern :-

While driving vehicle on road - Back door rattle noise & Sagging concern is detected.

While driving on high bumps on rattle path generates the noise.

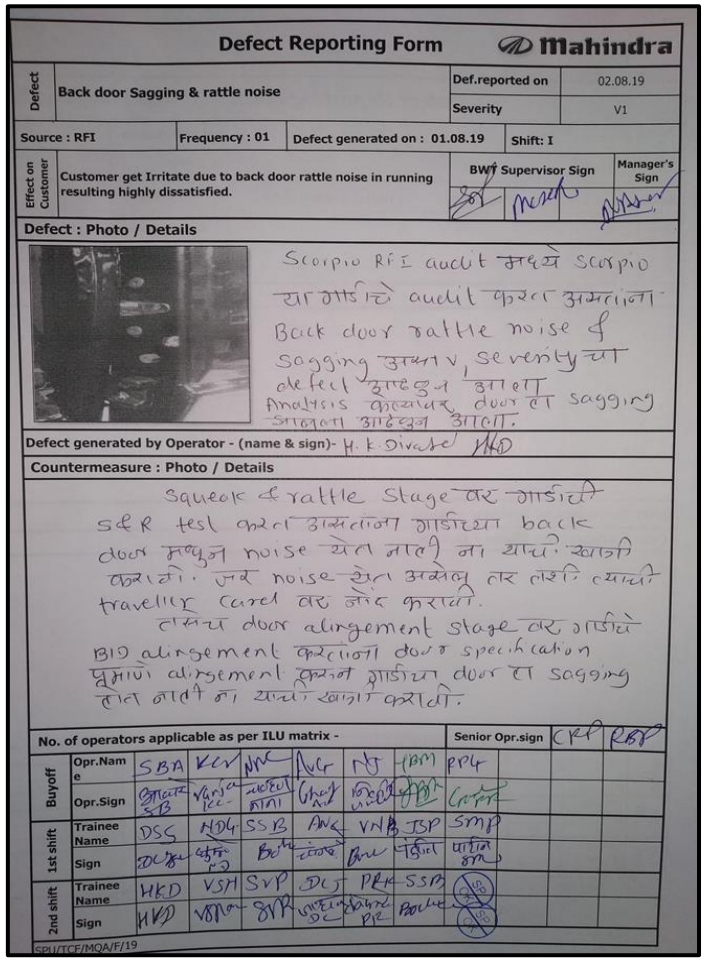
Due to this it may create issue of customer satisfaction while driving the vehicle.

Cost of Poor Quality-Rework

Rework Cost estimation for Back Door Abnormal Noise issue		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	5.0
Total Required Rework time per day	Min	75
Effective working time	Min	465
Operator required /day	nos	0.16
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	52.42
Avg no. of working days	days	25
Operator cost /month	Rs	1310.5
Operator Cost/year	**Rs**	**15725.8**

2.b Immediate Containment Action

- Awareness given for detection &100% Rework CAI.



Defect Reporting Form — Mahindra

Defect	Back door Sagging & rattle noise	Def.reported on	02.08.19
		Severity	V1

Source : RFI	Frequency : 01	Defect generated on : 01.08.19	Shift: I

Effect on Customer	Customer get Irritate due to back door rattle noise in running resulting highly dissatisfied.	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

Scorpio RFI audit मध्ये Scorpio या गाडीचे audit करत असताना Back door rattle noise & sagging असा V, Severity चा defect आढळून आला. Analysis करताना, door ला sagging असल्या आढळून आला.

Defect generated by Operator - (name & sign)- H. K. Divate HKD

Countermeasure : Photo / Details

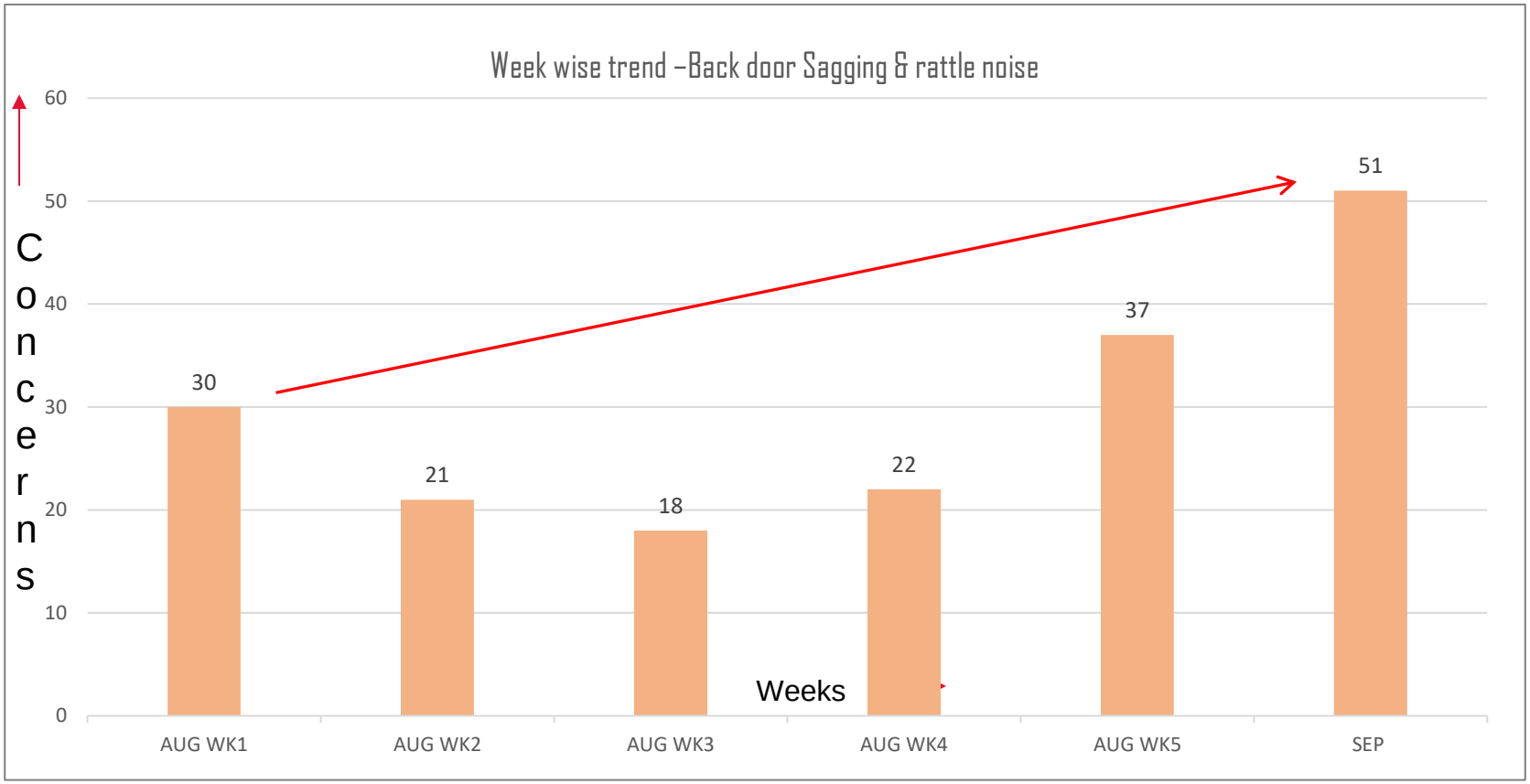
Squeak & rattle Stage वर गाडीची S&R test करत असताना गाडीचा back door मधून noise येत नाही ना याची खात्री करावी. जर noise येत असेल तर तशी त्याची traveller card वर नोंद करावी.

तसेच door alingement stage वर गाडीचे BIO alingement करताना door specification प्रमाणे alingement करून गाडीचा door ला sagging होत नाही ना याची खात्री करावी.

No. of operators applicable as per ILU matrix -									Senior Opr.sign	CRP	RBP
Buyoff	Opr.Name	SBA	KCV	NMC	NVG	NJ	HBM	PPG			
	Opr.Sign										
1st shift	Trainee Name	DSS	ND4	SSB	AWG	VNB	JSP	SMP			
	Sign										
2nd shift	Trainee Name	HKD	VSH	SVP	DCJ	PRA	SSB				
	Sign	HKD									

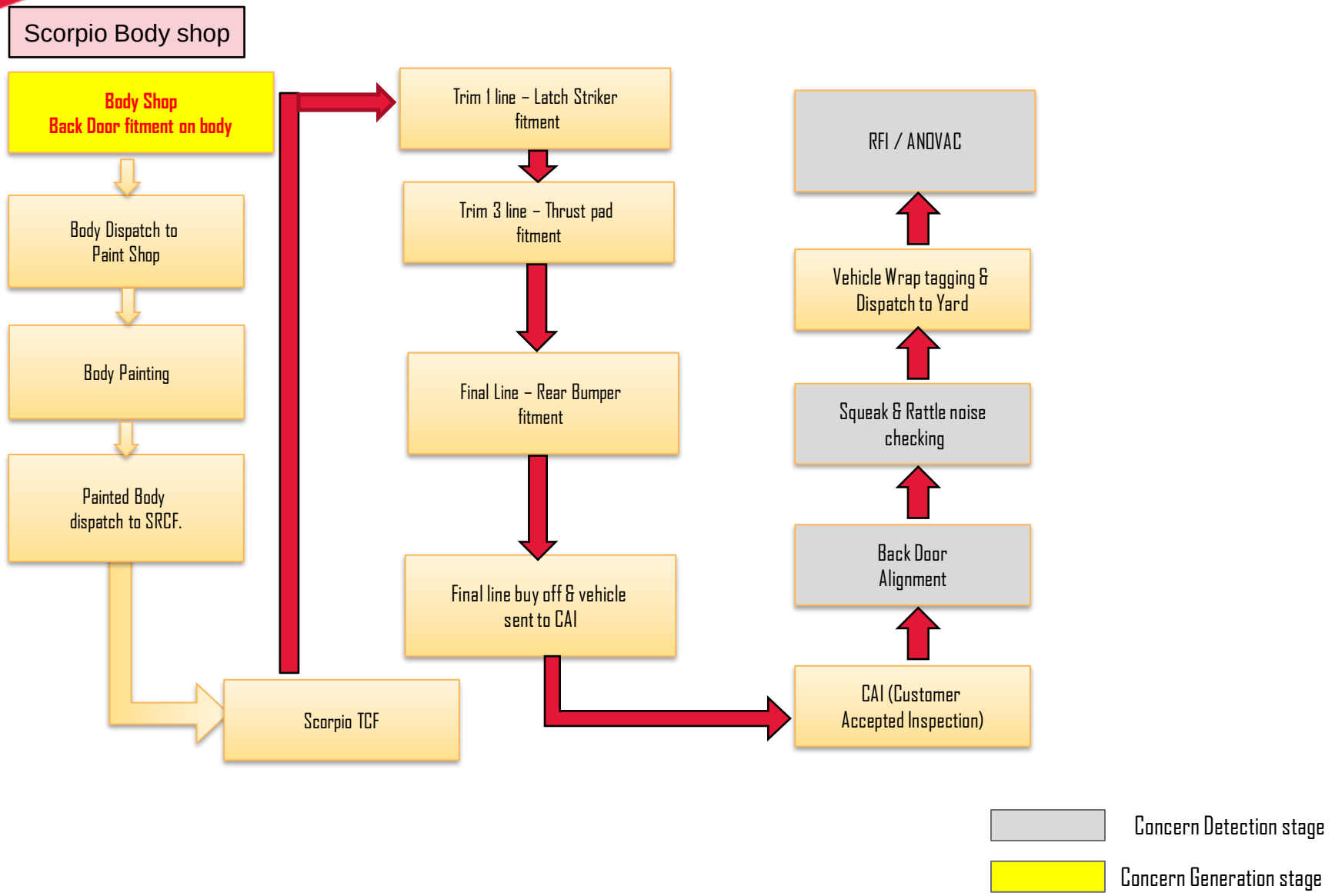
GPU/ICF/MQA/F/19

3. Understand the current situation



Inference – Defect rate increases Weekly, with Avg 166 incidence per month.

Step 3. Understanding Current Situation



Step 3. Understanding Current Situation

Mahindra — **4 M Analysis**

PU/ Area : Scorpio / Xylo TCF
Cell / Line : cai
BWT : 13

Requested By : MQA | Date -02.08.19 | Source - CAI | Conf. Analysis Sr.No :

Problem / Concern Statement :- Back door rattle noise & Sagging in Scorpio W105.

FMEA Severity - Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (I) :03 | Vehicle/ Aggre. No. & model - Scorpio W105 | Concern Priority Number :

Date of Incidence :04.04.19 | Shift / Time : 1st shift | Whether SC/CC/Z parameter affected ? ☐ YES ☐ NO ☐ Z ☐ CC ☑ SC

Immediate containment action for Customer Protection :
Yard check done.

Does a related Conformity Analysis or QC story already exist ? ☐ YES ☑ NO | If Yes | Conformity Analysis No | QC Story Sr. No-

Failure mode or cause/s identified -

Problem Assigned to - | BWT :

Stage / Post Analysis -

Does a control already exist ? ☑ YES ☐ NO

Man (Human actions)

Operator name -N N chavan
Category - ☑ DR ☐ N1 ☐ EPP ☐ APP
Is operator trained on the stage ? ☑ YES ☐ NO
Operator level (I.L.U) & Dexterity level(1..4) 4
Is operator under training ? ☐ under trg.on stage
☑ QC Stage ☐ Redeployed after 1 month ☐ 3*3 matrix completed
Bad rework ☐ YES ☑ NO Newly deployed for rework - Yes
Is the specified facility used ? ☑ YES ☐ NO
Last calibration dates 15.9.14
Facility in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☑ YES ☐ NO
Are machine parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Method (Process)

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP/CP/SOS ? ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the checkman/ Buy off list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☑ YES ☐ NO
Buyoff opr.name - S.P.Chittewan
In only one shift ? ☐ YES ☑ NO
In only one line ? ☑ YES ☐ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO

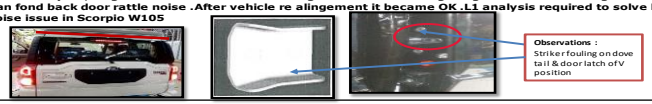
Others

Is there a SOS/ SOP / CP/monitoring plan ? ☑ YES ☐ NO
Is it implemented ? ☐ YES ☐ NO
Does it conform ? ☐ YES ☐ NO
Is the packaging in good condition ? ☐ YES ☐ NO
Last prevetive measures taken on -
Is there any alert mechanism for failure ? ☐ YES ☐ NO
Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☐ YES ☑ NO
Is the storage SOP respected ? ☐ YES ☐ NO
Is the shelf life / expiry date respected ? ☑ YES ☐ NO

Machine | Product Failure | Material / Consumable

Note : 1. If both fields are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
Durining traveller card analysis ,Stage associate N N Chavan having Dex.level -L3 doing alingement OK & follwing SOS .At rattle stage S.P.Chittewan fond back door rattle noise .After vehicle re alingement it became OK .L1 analysis required to solve back door back door rattle noise issue in Scorpio W105



Is Level 1 analysis required ? ☑ YES ☐ NO

1.Containment Action
1) DRF GIVEN TO STAGE ASSOCIATES .

Resp	Date	Status
SP	04.04.19	C

2. Permanent Action -
To be decided after detail analysis.

Resp	Date	Status
SP/Dv		O

Sustenance Plan for PCA (Audit, Pokayoke)-

Analysed by Name/s - Satish Patil | **Date -** 02.08.19

Choice : ☐ WSS ☑ QC Story ☐ Human Error Checksheet

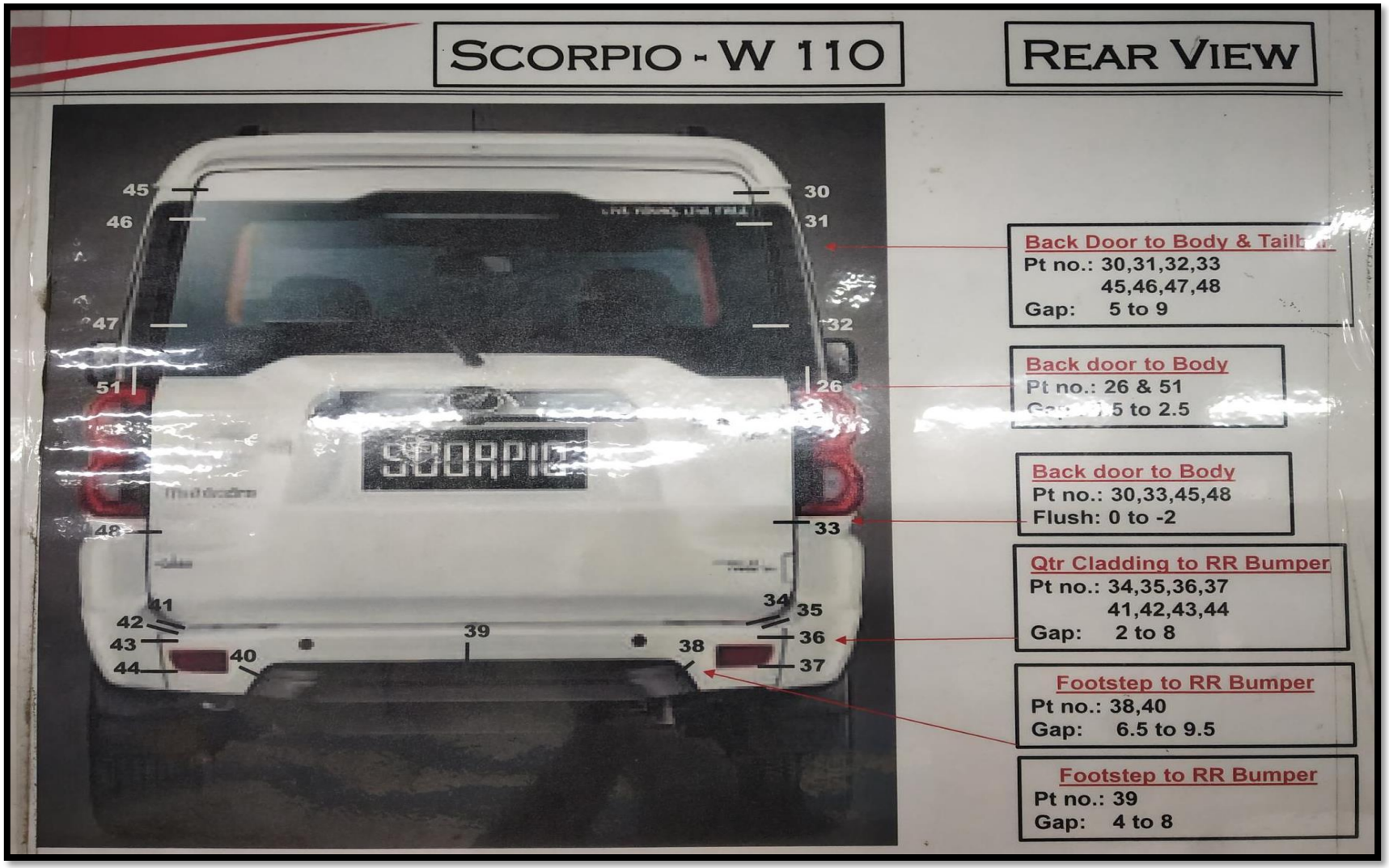
4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3
Method	Operator aware of key point and check present point
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows Vehicle & Part
Material	Handling and shelf life of part adhered

Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed only in Scorpio Models no issue in Marazzo.
- Associate is dexterity L3.
- Associate follows main step and key points.
- Associate is following his SOS while doing fitment Latch fitment and striker fitment
- Torque used for tightening is as per control plan and torque sheet.
- Door Alignment stage associates working as per sos.
- Concern observed in both shifts on Door alignment, Squeak & Rattle stage
- Concern observed in W110 Models only.

W110 Back Door Gap / Flush Issues Specifications :-

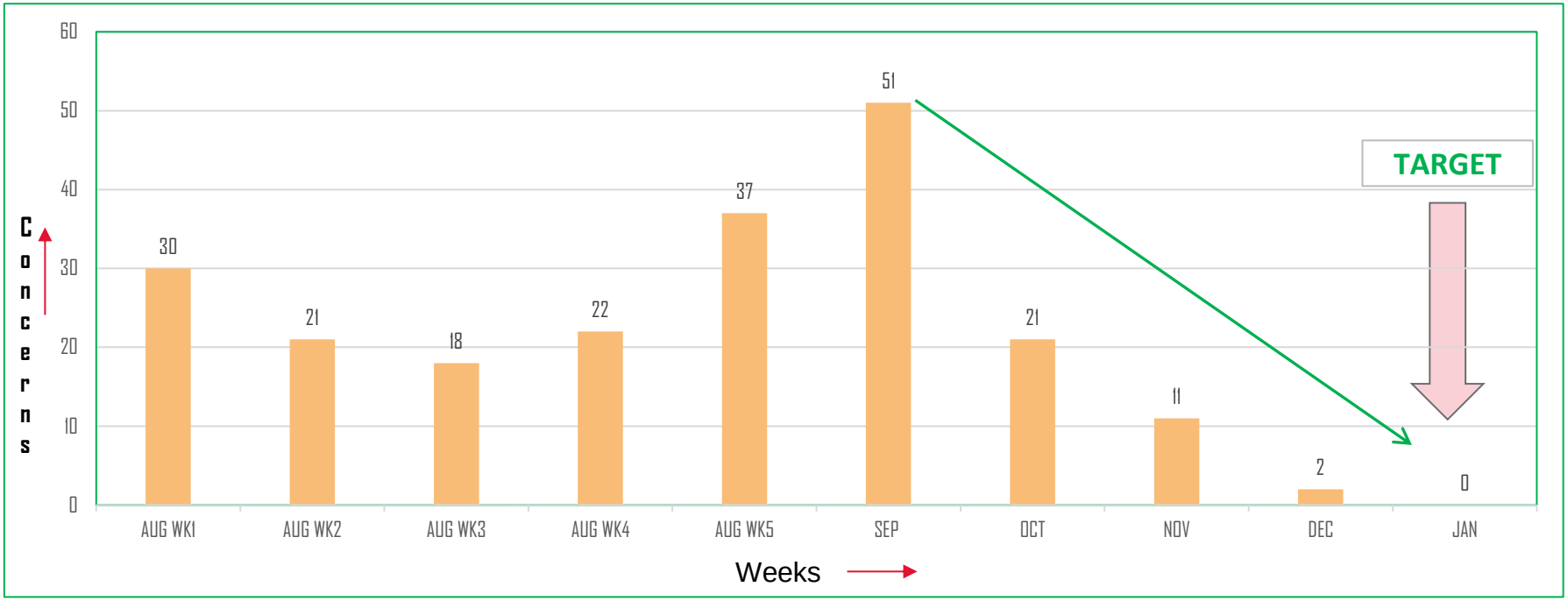


4. Select targets

Identify the target - target is a number indicating the level of improvement to be achieved.

Present Level: 166 cases reported at CAI based on data of last months

Target : To eliminate the concern of Back door Sagging & rattle noise issue in W110 Model by end of Dec -2019



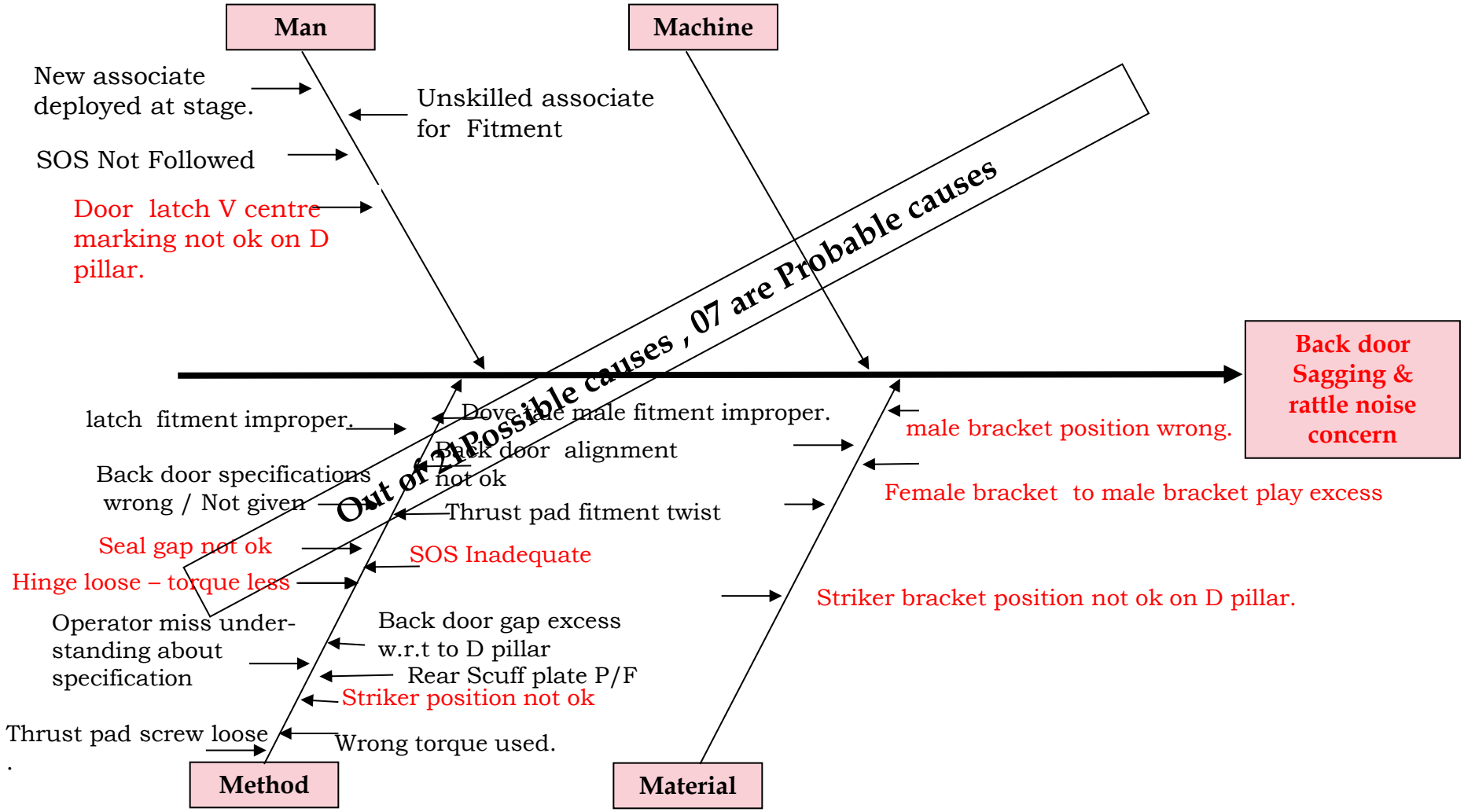
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	New associate deployed at stage.
2	SOS Not Followed .
3	Seal Gap Not OK
4	Unskilled associate for fitment.
5	Striker position not ok
6	Back door gap excess w.r.t to D pillar
7	Door latch V center marking not ok on D pillar.
8	Back door specifications wrong / Not given
9	Operator miss understanding about specification
10	Wrong torque used.
11	Back door alignment not ok
12	Latch fitment improper.
13	Dove tale male fitment improper.
14	Thrust pad fitment twist
15	Male bracket Position wrong.
16	Female bracket to male bracket play excess
17	Striker bracket position not ok on D pillar.
18	Hinge Loose-Torque less
19	Rear scuff plate poor fit
20	Thrust pad screw loose.
21	SOS in adequate.

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes identification in Brain storming session

No.	Probable Cause	Satish	Nitin	Prasad	Sachin	Vispute	Gobee	Jayesh	Total
1	New associate deployed at stage.	1	1	3	3	3	1	3	15
2	SOS Not Followed .	3	3	1	1	3	1	3	15
3	Seal Gap Not OK	3	3	9	3	9	9	9	45
4	Unskilled associate for fitment.	1	1	1	1	1	1	3	9
5	Striker position not ok	3	3	9	9	3	9	9	45
6	Back door gap excess w.r.t to D pillar	3	0	1	3	1	1	0	9
7	Door latch V center marking not ok on D pillar.	3	6	6	3	9	9	9	45
8	Back door specifications wrong / Not given	0	1	0	0	1	0	0	2
9	Operator miss understanding about specification	0	0	1	0	1	0	0	2
10	Wrong torque used.	3	3	3	0	3	0	0	12
11	Back door alignment not ok	1	0	1	0	3	0	1	6
12	Latch fitment improper.	3	3	3	0	0	0	6	15
13	Dove tale male fitment improper.	1	0	3	1	1	1	6	13
14	Thrust pad fitment twist	0	0	3	0	0	1	6	10
15	Male bracket Position wrong.	3	9	9	6	6	3	6	42
16	Female bracket to male bracket play excess	3	9	9	6	3	6	6	42
17	Striker bracket position not ok on D pillar.	6	6	9	3	6	6	9	45
18	Hinge Loose-Torque less	3	9	9	6	3	6	6	42
19	Rear scuff plate poor fit	0	1	0	0	1	0	0	2
20	Thrust pad screw loose.	3	3	3	0	3	0	0	12
21	SOS in adequate.	9	6	3	6	3	9	9	45

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact. **Threshold Value for selecting probable cause is 42.**

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

No.	Not considered Cause	Reasons
1	New associate deployed at stage.	All trained operator deployed on stage
2	SOS Not Followed .	All trained operator deployed on stage
3	Unskilled associate for fitment.	All trained operator deployed on stage
4	Back door gap excess w.r.t to D pillar	Gap observed within specifications
5	Back door specifications wrong / Not given	Specifications checked found ok
6	Operator miss understanding about specification	Check with operators found ok
7	Wrong torque used.	Gun torque checked with ACTA found ok
8	back door alignment not ok	Checked & found ok
9	latch fitment improper.	Checked & found ok
10	Dove tale male fitment improper.	Checked & found ok
11	Thrust pad fitment twist	Checked & found ok
12	Rear scuff plate poor fit	Checked & found ok
13	Thrust pad screw loose.	Checked & found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. no	Probable Causes
1	Seal gap not ok
2	Striker mounting not ok
3	Door latch V centre marking not ok on D pillar.
4	Male bracket position wrong.
5	Female bracket to male bracket play excess
6	Striker bracket position not ok on D pillar.
7	Hinge loose-torque less
8	SOS Inadequate

TEST OF HYPOTHESIS

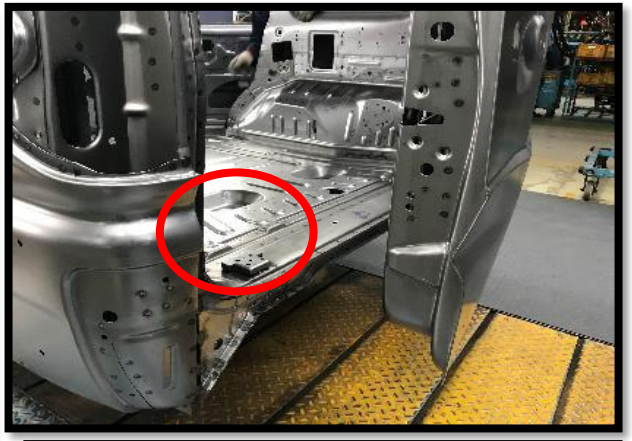
5.2 Confirm the probable cause:

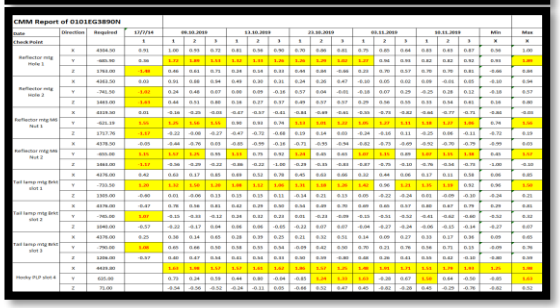
No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Seal gap not ok	Body shop back door fitment process verification for use of FRP fixture & checking of seal after door fitment with the help of Taper Scale.	50 vehicles audited at body shop 100% FRP fixture used for door fitment and door seal gap checked found sagging of 0.8 to 1.4 mm to Lh side. Spect – 5 to 9 mm	Hypothesis valid

Observations : -

After analysing 50 BIW at TCF fitment stage found sagging of 0.8 to 1.4 mm to Lh side.

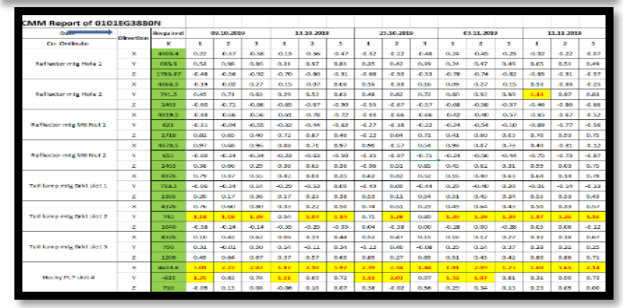
So this hypothesis is valid.





CMM Report of 0101EG3890N

Date			17/7/14	09.10.2019			11.10.2019			21.10.2019			01.11.2019			10.11.2019			Min	Max
Check Point		Direction	Required	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	X	X
Reflector mtg Hole 1		X	4304.50	0.91	1.00	0.93	0.72	0.81	0.98	0.90	0.70	0.86	0.81	0.75	0.65	0.64	0.83	0.83	0.87	0.98
		Y	-665.90	0.96	1.22	1.09	1.11	1.22	1.11	1.26	1.26	1.09	1.02	1.07	0.94	0.99	0.82	0.92	0.93	1.09
		Z	1793.00	1.09	0.86	0.81	0.75	0.94	0.94	0.93	0.84	0.86	-0.08	0.20	0.70	0.57	0.70	0.70	0.81	0.86	1.09
Reflector mtg Hole 2		X	4363.50	0.03	-0.91	0.88	0.94	0.49	0.30	0.31	0.24	0.20	0.47	-0.10	0.05	0.02	0.09	-0.01	0.05	-0.10	0.94
		Y	-741.50	-1.02	-0.24	0.46	0.07	0.00	0.09	-0.18	0.57	0.04	-0.01	-0.18	0.07	0.29	-0.25	0.28	0.12	-0.18	0.57
		Z	1483.00	-1.03	0.44	0.51	0.80	0.18	0.27	0.37	0.49	0.57	0.57	0.29	0.56	0.55	0.53	0.54	0.61	0.18	0.80
Reflector mtg M6 Nut 1		X	4419.50	0.01	-0.14	-0.25	-0.03	-0.47	-0.57	-0.41	-0.44	-0.49	-0.41	-0.35	-0.73	-0.42	-0.44	-0.77	-0.71	-0.80	-0.03
		Y	-621.19	1.15	1.25	1.15	1.15	0.90	0.92	0.74	1.13	1.05	1.12	1.05	1.27	1.11	1.18	1.27	1.05	0.74	1.30
		Z	1717.76	1.17	-0.22	-0.06	-0.27	-0.47	-0.72	-0.68	0.19	0.14	0.03	-0.24	-0.16	0.11	-0.25	0.06	-0.11	-0.72	0.19
Reflector mtg M6 Nut 2		X	4578.50	-0.05	-0.44	-0.70	0.03	-0.85	-0.99	-0.18	-0.71	-0.55	-0.54	-0.82	-0.75	-0.69	-0.52	-0.70	-0.75	-0.99	0.03
		Y	-655.00	1.15	1.17	1.15	0.98	1.11	0.75	0.92	1.14	0.85	0.85	1.07	1.15	0.85	1.07	1.15	1.18	0.85	1.17
		Z	1684.00	1.17	-0.36	-0.29	-0.22	-0.86	-0.22	-1.00	-0.29	-0.33	-0.83	-0.67	-0.79	-0.10	-0.76	-0.64	-0.79	-1.00	-0.10
Tail lamp mtg Bolt slot 1		X	4378.00	0.42	0.63	0.17	0.85	0.69	0.52	0.78	0.45	0.62	0.66	0.32	0.44	0.06	0.17	0.11	0.58	0.06	0.85
		Y	-731.50	1.20	1.12	1.10	1.20	1.08	1.12	1.06	1.15	1.10	1.20	1.02	0.96	1.21	1.15	1.20	0.92	0.96	1.30
		Z	1305.00	-0.80	0.01	-0.00	0.13	0.15	0.11	-0.14	0.21	0.15	0.05	-0.22	-0.24	0.01	-0.05	-0.10	-0.24	0.21	0.21
Tail lamp mtg Bolt slot 2		X	4376.00	-0.67	0.78	0.56	0.81	0.62	0.29	0.50	0.54	0.69	0.70	0.69	0.65	0.57	0.80	0.67	0.78	0.29	0.81
		Y	-765.00	1.07	-0.15	-0.33	-0.12	0.24	0.32	0.23	0.01	-0.23	-0.09	-0.15	-0.31	-0.32	-0.41	-0.62	-0.60	-0.32	0.32
		Z	1040.00	-0.57	-0.22	-0.17	0.04	0.06	0.06	-0.05	-0.22	0.07	0.07	-0.04	-0.27	-0.24	-0.06	-0.13	-0.14	-0.27	0.07
Tail lamp mtg Bolt slot 3		X	4378.00	0.25	0.38	0.14	0.65	0.28	0.39	0.25	0.21	0.32	0.31	0.14	0.09	0.27	0.33	0.17	0.38	0.09	0.65
		Y	-790.00	1.08	0.65	0.66	0.56	0.58	0.55	0.54	-0.09	0.42	0.50	0.70	0.21	0.76	0.56	0.71	0.15	-0.09	0.76
		Z	1206.00	-0.57	0.40	0.47	0.54	0.41	0.54	0.43	0.50	0.59	0.60	0.46	0.26	0.41	0.55	0.62	0.40	-0.60	0.59
Hooky PLP slot 4		X	4429.00		1.07	1.08	1.17	1.17	1.07	1.17	1.00	1.17	1.20	1.00	1.00	1.23	1.10	1.23	1.00	1.20	1.08
		Y	615.00		0.75	0.24	0.59	0.84	0.80	-0.04	-0.15	1.07	1.10	1.00	-0.28	0.67	1.10	0.84	-0.10	-0.05	1.07
		Z	70.00		-0.54	-0.56	-0.52	-0.24	-0.11	0.05	-0.66	-0.52	0.47	0.45	-0.02	-0.26	0.45	-0.29	-0.76	-0.02	0.52



CMM Report of 0101EG3880N

Date		Required	09.10.2019			23.10.2019			25.10.2019			01.11.2019			11.11.2019		
Co Ordinate	Direction	X	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
Reflector mtg Hole 1	X	4306.4	0.22	0.57	0.58	0.13	-0.96	-0.47	-0.12	-0.22	-0.48	-0.24	-0.45	-0.26	-0.32	-0.22	-0.57
	Y	665.4	0.53	0.98	0.85	0.31	0.47	0.81	0.25	0.42	0.36	0.34	0.47	0.45	0.65	0.51	0.46
	Z	1793.87	-0.48	-0.58	-0.90	-0.70	-0.80	-0.91	-0.88	-0.53	-0.53	-0.78	-0.74	-0.82	-0.85	-0.91	-0.57
Reflector mtg Hole 2	X	4364.5	0.19	0.02	0.27	0.15	-0.97	0.81	0.56	0.48	0.56	0.09	0.27	0.15	0.54	-0.95	0.25
	Y	741.5	0.45	0.74	0.63	0.29	0.53	0.41	0.48	0.82	0.70	0.60	0.93	0.95	1.03	0.97	0.83
	Z	1483	-0.60	-0.71	-0.86	-0.65	-0.97	-0.90	-0.55	-0.67	-0.57	-0.68	-0.58	-0.57	-0.46	-0.86	-0.88
Reflector mtg M6 Nut 1	X	4419.5	-0.88	-0.66	-0.36	-0.91	-0.76	-0.72	-0.56	-0.66	-0.66	-0.42	-0.90	-0.57	-0.95	-0.97	-0.52
	Y	621	-0.21	-0.04	-0.05	-0.32	-0.96	-0.82	-0.27	-0.18	-0.22	-0.24	-0.04	-0.16	-0.99	-0.77	-0.59
	Z	1718	0.82	0.60	0.40	0.72	0.87	0.46	-0.12	0.64	0.71	0.41	0.80	0.83	0.70	0.53	0.75
Reflector mtg M6 Nut 2	X	4578.5	0.97	0.68	0.96	0.88	0.71	0.92	0.96	-0.12	0.54	0.96	0.82	0.70	0.40	-0.81	-0.52
	Y	655	-0.68	-0.24	-0.24	-0.22	-0.92	-0.59	-0.93	-0.97	-0.71	-0.24	-0.06	-0.96	-0.75	-0.73	-0.87
	Z	1683	0.28	0.06	0.27	0.20	0.62	0.21	-0.09	0.03	0.40	0.42	0.62	0.31	0.77	0.63	0.75
Tail lamp mtg Bolt slot 1	X	4376	0.79	0.97	0.50	0.42	0.69	0.25	0.62	0.52	0.50	0.50	0.80	0.63	0.69	0.59	0.78
	Y	731.5	-0.06	-0.24	0.14	-0.20	-0.52	0.03	-0.19	0.00	-0.44	0.20	-0.90	0.20	-0.21	-0.25	-0.12
	Z	1305	0.28	0.17	0.20	0.17	0.22	0.28	0.03	0.13	0.04	0.13	0.40	0.20	0.33	0.33	0.42
Tail lamp mtg Bolt slot 2	X	4376	0.76	0.60	0.80	0.83	0.22	0.50	0.74	0.53	0.23	0.49	0.46	0.83	0.50	0.33	0.57
	Y	765	1.18	1.19	1.20	0.91	1.03	1.03	0.75	1.28	0.80	1.20	1.20	1.20	1.17	1.26	1.10
	Z	1040	-0.18	-0.24	-0.14	-0.30	-0.25	-0.23	0.04	-0.18	0.00	-0.28	0.00	-0.28	0.03	0.08	-0.12
Tail lamp mtg Bolt slot 3	X	4376	0.18	0.60	0.62	0.86	0.39	0.46	0.52	0.67	0.55	0.56	0.57	0.27	0.52	0.36	0.67
	Y	790	0.31	-0.01	0.50	0.24	-0.11	0.24	-0.12	0.46	-0.08	0.20	0.14	0.27	0.23	0.21	0.25
	Z	1206	0.48	0.64	0.67	0.37	0.57	0.43	0.68	0.27	0.68	0.51	0.48	0.42	0.80	0.68	0.71
Hooky PLP slot 4	X	4429.4	1.05	0.97	0.97	0.93	0.90	0.97	0.95	0.96	1.00	1.03	0.97	1.03	0.90	1.03	0.98
	Y	615	1.25	0.80	0.76	1.13	0.83	0.72	1.03	1.03	0.97	1.10	1.07	0.84	0.24	0.88	0.73
	Z	70	-0.08	0.13	0.50	-0.08	0.10	0.47	0.38	-0.02	0.56	0.20	0.34	0.13	0.23	0.68	0.60

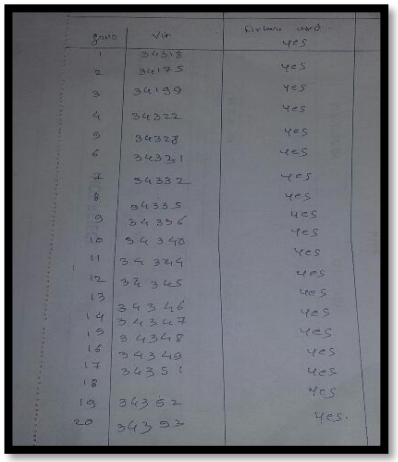
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Striker mounting not ok	Visual inspection of Striker fitment process at trim-1 line. Confirm whether stage associate is using fixture for striker setting or not.	20 vehicles checked and no discrepancy observed. Point already available in MQA process Audit – Records checked no NC found in last 3 month's data.	Hypothesis invalid

Observations : -

Fixture used on line for striker fitment for 100% cases .
so it is not a valid cause.



Sno	Vin	Fixture used
1	34338	yes
2	34175	yes
3	34199	yes
4	34322	yes
5	34328	yes
6	34331	yes
7	34332	yes
8	34335	yes
9	34336	yes
10	34340	yes
11	34344	yes
12	34345	yes
13	34346	yes
14	34347	yes
15	34348	yes
16	34349	yes
17	34351	yes
18		yes
19	34352	yes
20	34353	yes.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Door latch V center marking not ok on D pillar.	Door latch marking done on D pillar by Pencil marking i.e. Scribing	Door latch V position shifted to -0.8 mm in Z direction	Hypothesis valid

Observations :

Variation observed in Door latch V position shifted to -0.8 mm in Z direction there for it's a valid cause hence this hypothesis is valid.

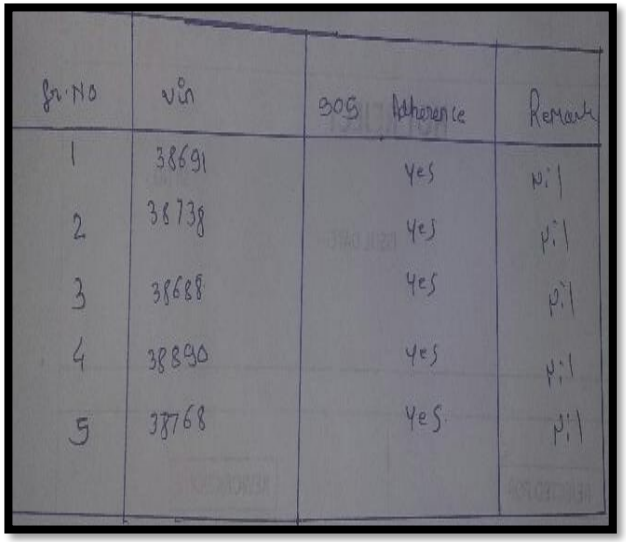
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Male bracket position wrong	Door panel to be checked on gauge	5 Door panels checked on gauge all hole positions found Ok as per specifications	Hypothesis invalid

Observations :

- 5 Samples of door panel checked on Gauge no Discrepancy observed.

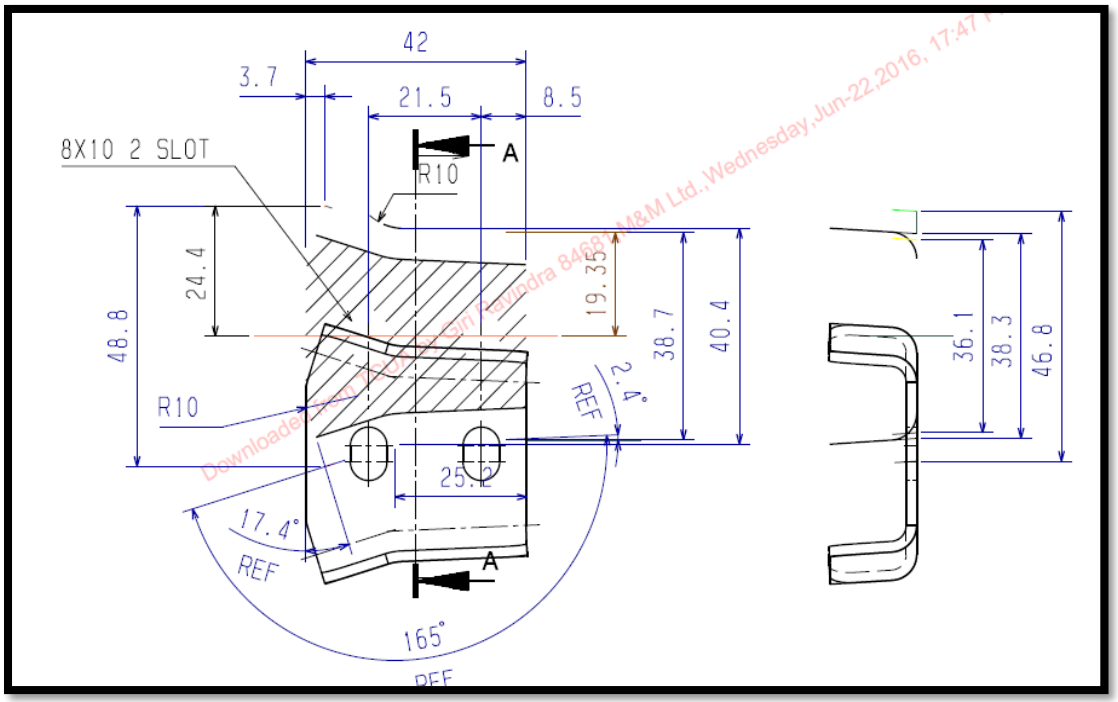


Sr.No	VIN	30S	Adherence	Remark
1	38691		Yes	Nil
2	38738		Yes	Nil
3	38688		Yes	Nil
4	38890		Yes	Nil
5	38768		Yes	Nil

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Female bracket to male bracket play excess	Dimensional checking	14 parts of dove tail checked , All parts dove tale female opening found Ok with specifications. Spect. – 40.3 ± 0.3 mm	Hypothesis invalid



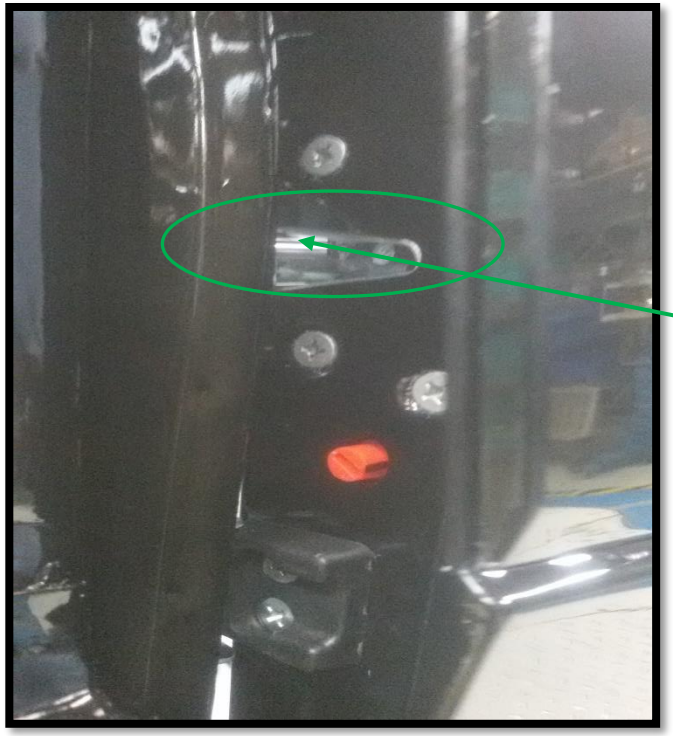
Observations :

- No variation observed in Female bracket inner opening hence it is invalid cause.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
6	Striker bracket position not ok on D pillar.	Striker bracket mounting position on body to be check with the help of CMM.	Striker mtg square slot and female part mtg hole position checked on CMM found OK with Specification in Z direction.	Hypothesis Invalid



Observations :
Fixture used on line for striker fitment for 100% cases so it is not a valid.
hence this hypothesis is Invalid.

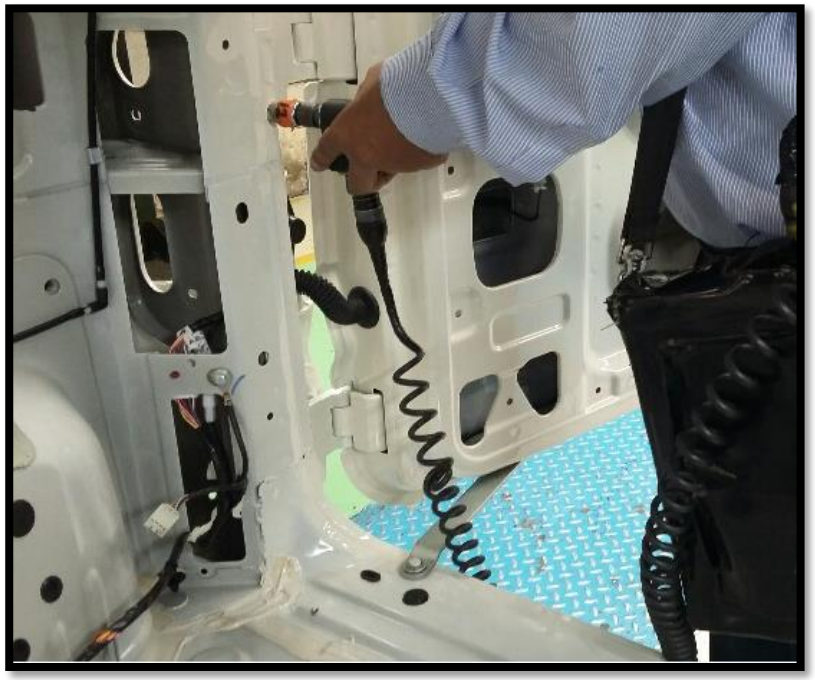
TEST OF HYPOTHESIS

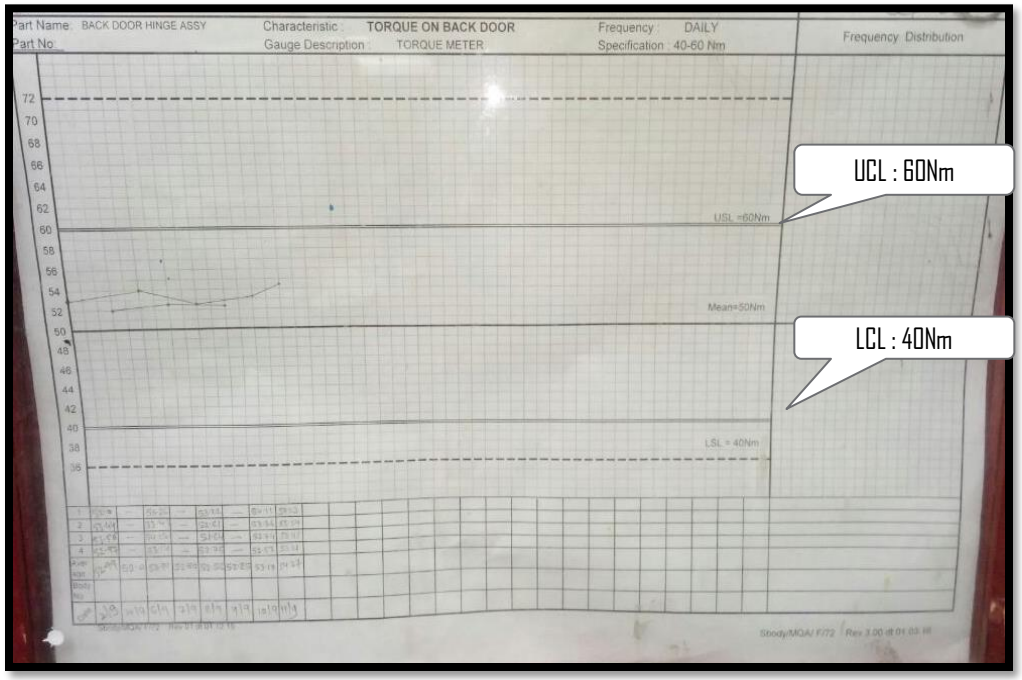
5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Hinge loose – torque less	Defective vehicles residual torque checking with the Torque Inspection Unit ACTA	15 Defective Vehicles residual torque checked all found within spec of 50±10Nm.	Hypothesis invalid

Observations :

15 vehicles audited and observed no discrepancy.





TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
8	SOS in adequate (Block Used).	Visual inspection for Alignment rework process & verification of same vehicles on squeak & Rattle stage for sticker fouling.	35 cycles observed and checked for sticker fouling on squeak & Rattle track . 7 vehicles found defective for sticker fouling for back door.	Hypothesis valid

No.	VIN No.	Block Used for B/D Sticker Setting		Back door Fouling After S&R	Remark
		Yes	No		
2	MA1TA2SJXH2H28271		No		
3	MA1TA2SJXH2H28978	`Yes		`Yes	B/D Fouling on Sticker
4	MA1TA2SJXH2H29125		No		
5	MA1TA2SJXH2H29149		No		
6	MA1TA2SJXH2H29182		No		
7	MA1TA2SJXH2H29192		No	`Yes	B/D Fouling on Sticker
8	MA1TA2SJXH2H29274		No		
9	MA1TA2SJXH2H29317		No		
10	MA1TA2SJXH2H29509	`Yes			
11	MA1TA2SJXH2H29581		No		
12	MA1TA2SJXH2H29843		No		
13	MA1TA2SJXH2H29843		No	`Yes	B/D Fouling on Sticker
14	MA1TA2SJXH2H30051		No		
15	MA1TA2SJXH2H30063		No		
16	MA1TA2SJXH2H30197		No		
17	MA1TA2TDKH2H28948	`Yes		`Yes	B/D Fouling on Sticker
18	MA1TA2TDKH2H29134	`Yes		`Yes	B/D Fouling on Sticker
19	MA1TA2TDKH2H29333		No		
20	MA1TA2TDKH2H29378		No		
21	MA1TA2TDKH2H29585		No		
22	MA1TA2TDKH2H29620		No		
23	MA1TA2TDKH2H29725	`Yes			
24	MA1TA2TDKH2H30140		No		
25	MA1TA2TDKH2H30307		No	`Yes	B/D Fouling on Sticker
26	MA1TA2TDKH2H30318		No		
27	MA1TA2WGXH2H29299	`Yes			
28	MA1TA2WGXH2H29300		No		
29	MA1TA2WGXH2H29717		No		
30	MA1TA2WGXH2H29730		No		
31	MA1TA2WGXH2H30216		No		
32	MA1TA2WGXH2H30297		No		
33	MA1TA2WLXH2H29676		No		
34	MA1TA4SJXH2H29043	`Yes		`Yes	B/D Fouling on Sticker
35	MA1TA4SJXH2H29610		No		

Observations :

- 35 vehicles checked on door alignment rework stage for which SOS correctly followed but still 07 vehicles reported on squeak and rattle stage for back door sticker fouling.
- Out of 7 defective vehicles stage associate has used PVC block for 4 vehicles to make door alignment in center of striker.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:-

No.	Probable Cause	Testing Method	Conclusion
1	Seal Gap Not OK	Dimensional Inspection	Hypothesis valid
2	Striker position not ok	Visual Inspection	Hypothesis Invalid
3	Door latch V center marking not ok on D pillar.	Visual Inspection	Hypothesis valid
4	Male bracket Position wrong.	Dimensional Inspection	Hypothesis Invalid
5	Female bracket to male bracket play excess	Visual Inspection	Hypothesis Invalid
6	Striker bracket position not ok on D pillar.	CMM Inspection	Hypothesis Invalid
7	Hinge loose-torque less	Visual Inspection	Hypothesis Invalid
8	SOS in adequate.	Visual Inspection	Hypothesis valid

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No.	Probable cause	WHY?	WHY?	WHY?	WHY?	Action
1	Seal Gap Not OK	Variation observed in Door found sagging of 0.8 to 1.4 mm to Lh side	No Proper profile spacer & pin locator for maintaing gaps			Compensation at seal gap to be add . Spacer & Pin locator to be develop.
2	Door latch V center marking not ok on D pillar.	Variation observed in Door latch V position shifted to -0.8 mm in Z	Qtr inner lower assly fixture position found shifted in -Z direction by 0.8.	Tolerance stack up issue was not considered during fixture design of Qtr inner assly.		Qtr inner lower assly fixture position to be shifted in -Z direction by 0.8 mm
3	SOS Inadequate (Block used)	SOS followed (Block used) by stage associate but still back door noise fouling reported at squeak & rattle stage.	For many vehicles during door alignment rework block used by associate to set back door position	Block used to minimize the variation between male & female striker without opening any part.	Use of block for door setting is the action decided in past for easy setting of door which inappropri ate.	SOS to be modified for not to use block for door setting & do setting by opening Male / female bkt as per req.

6. Implement the corrective measures

Possible Remedies Identified

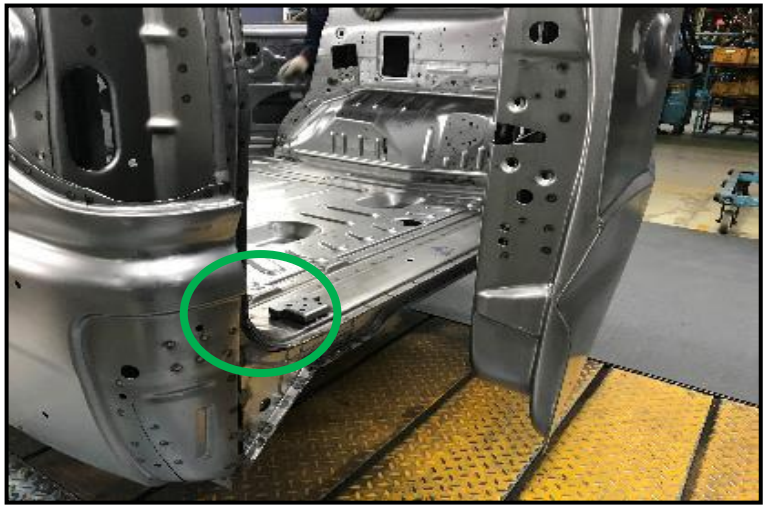
Sr No.	Probable cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Seal Gap Not OK	Compensation at seal gap to be added . Spacer & Pin locator to be develop.	High	Low	Yes	Low	Accepted
2	Door latch V center marking not ok on D pillar.	Qtr inner lower assly fixture position to be shifted in -Z direction (downward direction) by 0.8 mm at Qtr inner assy & after action striker found on door latch of V position in lower side upto 0.8 mm observed	High	Low	Yes	Low	Accepted
3	SOS inadequate (Block used)	SOS To be modified for not to use puller for door setting & do setting by opening Male / female bkt as per req.	Low	Low	Yes	Low	Accepted

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Seal Gap Not OK	Compensation of 1 mm added at seal gap . Developing profile spacers for maintaining gaps & New fixture for Rh critical gap. Also develop Pin locator as against profile	22.11.19	Nitin/pravin	Closed

After analysing 185 BIW at TCF fitment stage found sagging of 0.8 to 1.4 mm to Lh side



Compensation of 1 mm added at seal gap and same checked by No-Go gauge at body shop

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Seal Gap Not OK	Compensation of 1 mm added at seal gap . Developing profile spacers for maintaining gaps & New fixture for Rh critical gap. Also develop Pin locator as against profile	04.12.19	Nitin/Pravin	Closed

Before

Profile spacers for maintaining gaps

After

New fixture for Rh critical gap

Pin locator as against profile

In-house developed fixture

Total Improvements done- 3

Total Iterations done - 8

Total BIW checks - 1535

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Seal Gap Not OK	Compensation of 1 mm added at seal gap . Developing profile spacers for maintaining gaps & New fixture for Rh critical gap. Also develop Pin locator as against profile	22.11.19	Nitin/Pravin	Closed

			Band Width	
	Rh	Lh	Rh	Lh
R & D specification -	5.2 to 7.2 mm	5.2 to 7.2 mm		
F-19 Prior Reading-	4.5 to 8.5mm	8 to 10.5mm.	4mm	2.5mm
F-19 BIW	3.5 to 6.5mm	8 to 11.5mm	3mm	3.5mm
F20 Actual Reading-	4.5 to 8.0mm	7.5 to 10.5mm.	3.5mm	3mm
F20 BIW	4.5 to 6.5mm	8.5 to 11mm	2mm	2.5mm
Current Readings:-	5 to 8mm	7.5 to 10mm.	3mm	2.5mm
BIW current readings	5 to 6.5mm	8 to 10mm	1.5mm	2mm

CMM Report of 0101EG3SS0N

Cu. Ordinate	Direction	Required	09.10.2019 1	2	3	23.10.2019 1	2	3	21.10.2019 1	2	3	03.11.2019 1	2	3	11.11.2019 1	2	3
Reflector mtg Hole 1	X	4005.4	0.23	-0.57	-0.58	-0.13	-0.96	-0.47	-0.32	-0.22	-0.48	0.24	-0.45	-0.25	-0.32	-0.22	-0.87
	Y	685.6	0.53	0.98	0.85	0.31	0.97	0.81	0.35	0.42	0.36	0.34	0.47	0.45	0.45	0.51	0.49
	Z	1763.87	-0.48	-0.58	-0.93	-0.70	-0.90	-0.91	-0.88	-0.53	-0.53	-0.78	-0.74	-0.82	-0.85	-0.91	-0.57
Reflector mtg Hole 2	X	4004.3	-0.19	-0.02	0.27	0.13	-0.07	0.06	0.16	-0.08	0.16	0.09	0.27	0.15	0.53	-0.08	-0.23
	Y	741.5	0.45	0.71	0.61	0.39	0.53	0.61	0.48	0.60	0.73	0.60	0.61	0.60	1.03	0.67	0.69
	Z	3403	-0.68	-0.71	-0.86	-0.85	-0.97	-0.90	-0.55	-0.67	-0.57	-0.68	-0.58	-0.57	-0.46	-0.86	-0.88
Reflector mtg M6 Nut 1	X	4019.5	-0.48	0.06	0.06	0.00	-0.78	-0.72	-0.88	-0.66	0.66	0.62	0.40	-0.57	-0.50	-0.57	-0.52
	Y	621	-0.31	-0.04	-0.55	-0.32	-0.44	-0.62	-0.27	-0.18	-0.22	-0.24	-0.54	-0.10	-0.99	-0.77	-0.56
	Z	2718	0.82	0.95	0.90	0.72	0.87	0.48	-0.12	0.64	0.71	0.41	0.80	0.83	0.70	0.53	0.75
Reflector mtg M6 Nut 2	X	4025.5	0.97	0.68	0.90	0.86	0.71	0.57	0.98	-0.17	0.04	0.90	0.87	0.75	0.40	-0.91	-0.12
	Y	655	-0.08	-0.24	-0.24	-0.23	-0.59	-0.50	-0.25	-0.37	-0.71	-0.24	-0.56	-0.44	-0.75	-0.79	-0.87
	Z	2903	0.38	0.90	0.29	0.39	0.60	0.39	-0.39	0.02	0.80	0.40	0.82	0.31	0.59	0.60	0.73
Tail lamp mtg Bolt slot 1	X	4026	0.79	0.97	0.90	0.82	0.68	0.20	0.42	0.02	0.02	0.10	0.90	0.61	0.64	0.34	0.79
	Y	723.5	-0.06	-0.24	0.14	-0.29	-0.19	0.03	-0.93	0.05	-0.49	0.20	-0.40	0.20	-0.31	-0.24	-0.13
	Z	2305	0.20	0.17	0.90	0.17	0.23	0.38	0.53	0.13	0.04	0.31	0.40	0.24	0.53	0.53	0.43
Tail lamp mtg Bolt slot 2	X	4026	0.76	0.90	0.80	0.83	0.22	0.54	0.98	0.03	0.03	0.89	0.86	0.83	0.50	0.33	0.57
	Y	741	1.18	1.19	1.29	0.94	1.03	1.34	0.75	1.29	0.60	1.20	1.29	1.27	1.17	1.25	1.34
	Z	2340	-0.38	-0.24	-0.14	-0.30	-0.29	-0.39	0.04	-0.38	0.00	-0.28	-0.30	-0.28	0.03	0.08	-0.13
Tail lamp mtg Bolt slot 3	X	4026	0.10	0.40	0.60	0.06	0.19	0.48	0.52	0.67	0.15	0.16	0.17	0.27	0.32	0.16	0.67
	Y	720	0.21	-0.01	0.50	0.16	-0.11	0.34	-0.12	0.60	-0.08	0.20	0.16	0.27	0.29	0.21	0.25
	Z	2206	0.48	0.84	0.97	0.37	0.57	0.65	0.85	0.27	0.85	0.31	0.45	0.42	0.80	0.88	0.73
Hocky PLT slot 4	X	4019.8	1.03	1.20	1.02	1.07	1.30	1.07	1.19	1.08	1.06	1.09	1.20	1.23	1.00	1.05	1.16
	Y	-633	1.25	0.83	0.74	1.16	0.69	0.72	1.33	1.03	0.97	1.32	1.07	0.81	0.31	0.50	0.73
	Z	720	-0.09	0.13	0.06	-0.06	0.10	0.07	0.38	-0.02	0.56	0.25	0.34	0.13	0.23	0.05	0.06

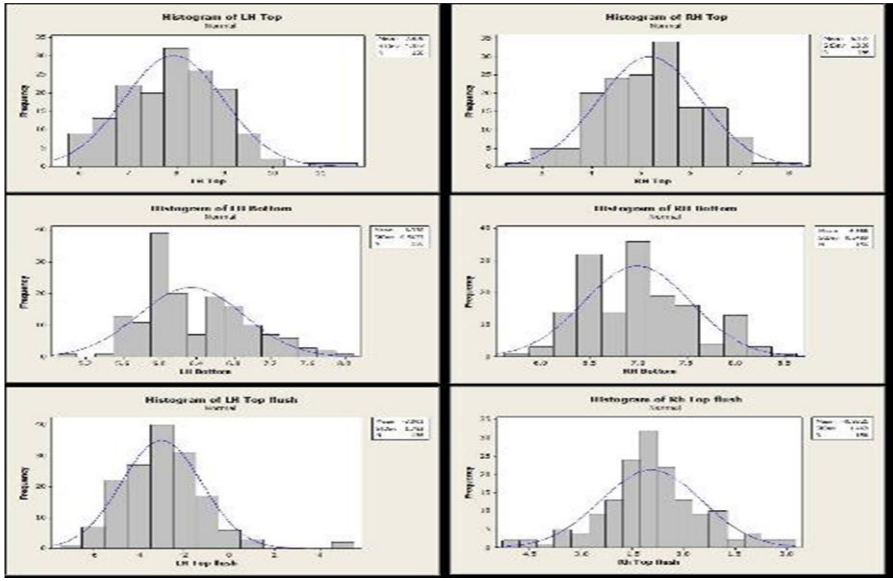
6. Implement the corrective measures

6.1 Actions Decided

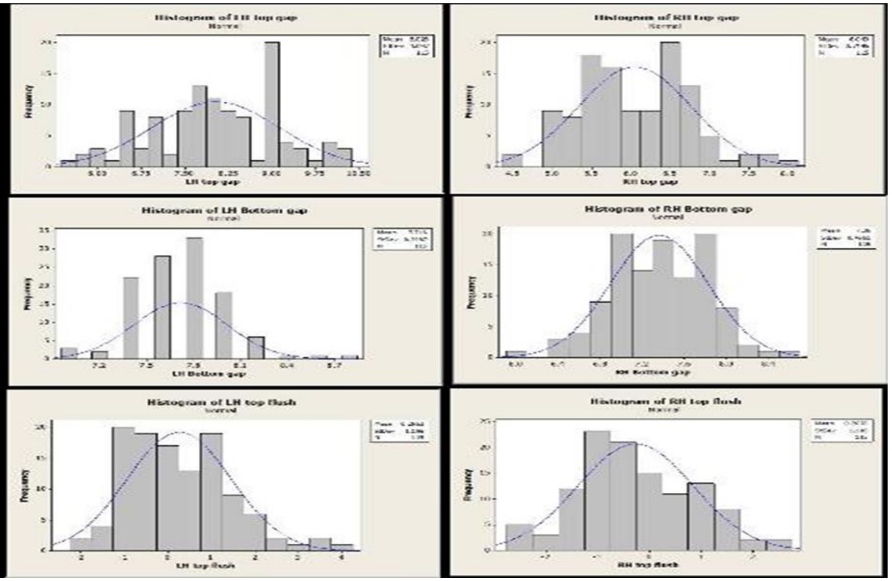
Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Seal Gap Not OK	Compensation of 1 mm added at seal gap . Developing profile spacers for maintaining gaps & New fixture for Rh critical gap. Also develop Pin locator as against profile	07.12.19	Nitin/Pravin	Closed

- **Back Door Gap & flush data are check on 185 Bodies in Scorpio Body shop.**
- **Same Vehicles was monitor for Gap & flush in TCF for Back Door.**
- **Data Analysis done with histograms.**

Body shop Readings



TCF Readings



6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2	Door latch V center marking not ok on D pillar.	Variation observed in Door latch V position shifted to -0.8 mm in Z	24.11.19	Nitin/Vispute	Closed

Qtr inner assy. found in Z direction by -0.8 mm

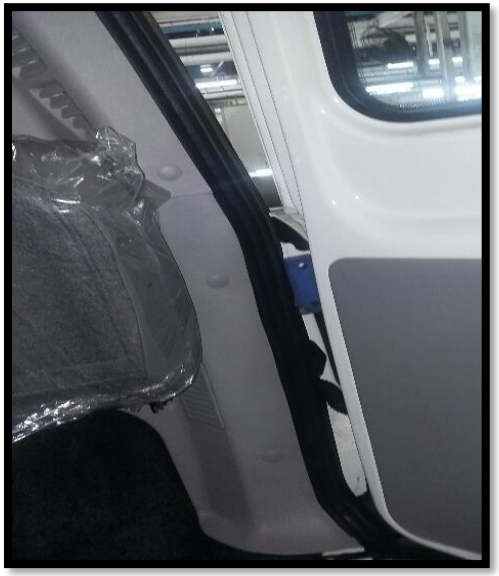
Qtr inner lower assly fixture position to be shifted in -Z direction (downward direction) by 0.8 mm at Qtr inner assy & after action Striker found on door latch of V position In lower side upto 0.8 mm observed – After action CMM done found OK

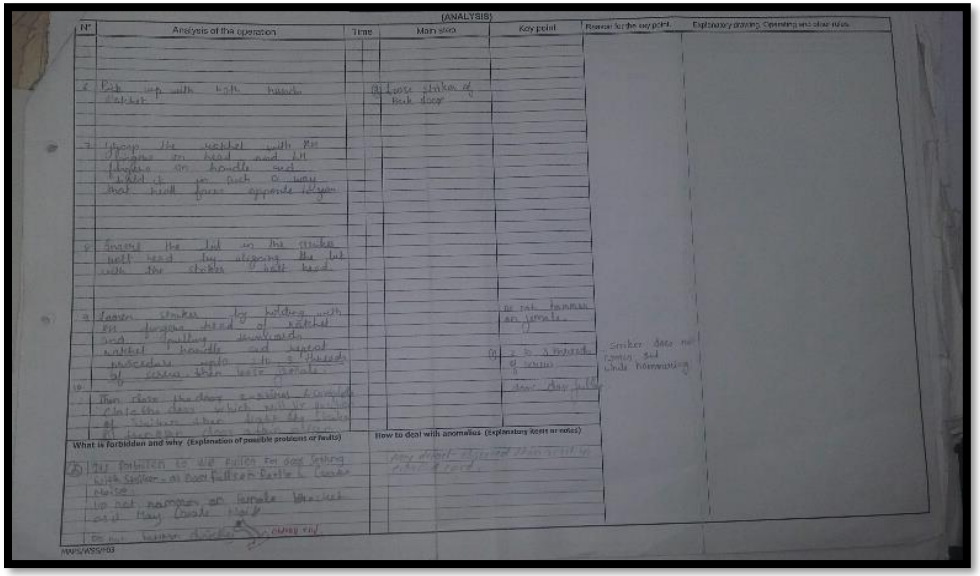
6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
3	SOS inadequate (Block used)	SOS modified by adding forbidden not to used block.	22.11.19	Satish Patil	Closed

Action : Forbidden point added in SOS for not to use block to lift door and alignment while rework.



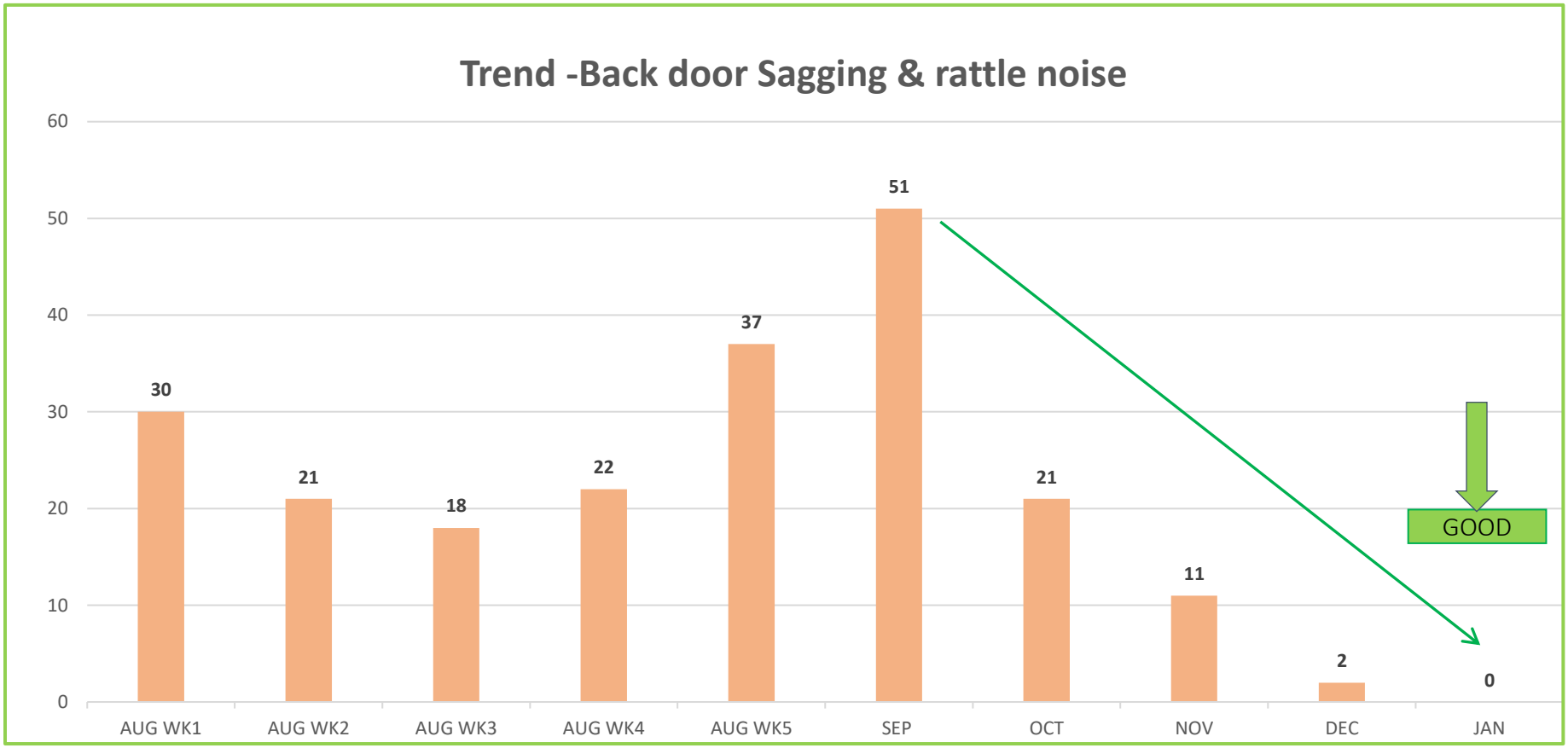


Block used in hinge for back door alignment rework

Sos updated – not to use block for alignment rework

7. Check & confirm the results & effects

7.1 Verify Results:



Inference –

02 Month data monitored after taking actions , 100 reduction achieved & actions are effective & sustained.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:-

S N	Action Taken	Actual date	Responsibility	Status
1	Compensation of 1 mm added at seal gap.	22.11.19	Nitin/Pravin	Closed
2	Fixture history card updated	15.01.19	Nitin/Pravin	Closed
3	Daily Supervisory Audit Check sheet updated.	13.01.19	Satish Patil	Closed

Check & confirm the results & effects (Cont.....)

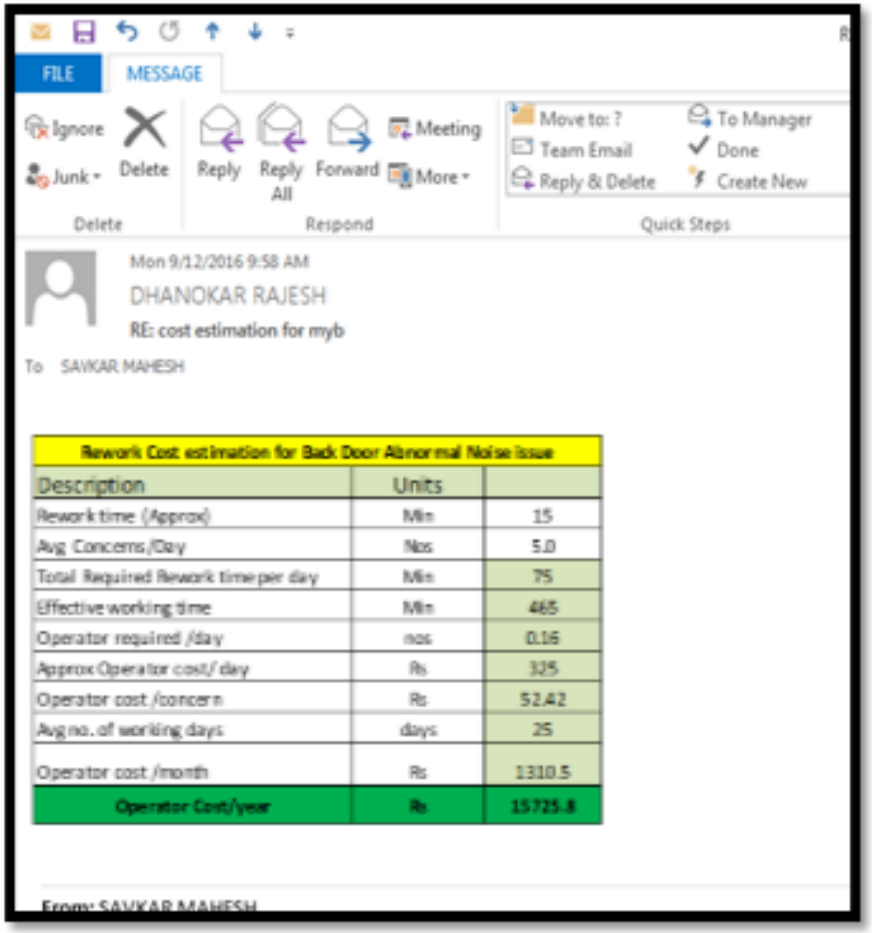
7.3 Tangible & Intangible benefits:

Tangible Benefits -

- More than 95% reduction in back door Sagging & Rattle noise issues.
- Rework cost is reduced by approx.Rs.15725/Annum.
- Quality indicators improved.
- Customer satisfaction (RFI , ANOVAC & Field)
- Through put time improved.

In tangible Benefits -

- Door alignment & Rattle associate moral improved.
- Line flow become smooth due to no rework



Rework Cost estimation for Back Door Abnormal Noise issue		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	5.0
Total Required Rework time per day	Min	75
Effective working time	Min	465
Operator required /day	nos	0.16
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	52.42
Avg no. of working days	days	25
Operator cost /month	Rs	1310.5
Operator Cost/year	Rs	15725.8

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

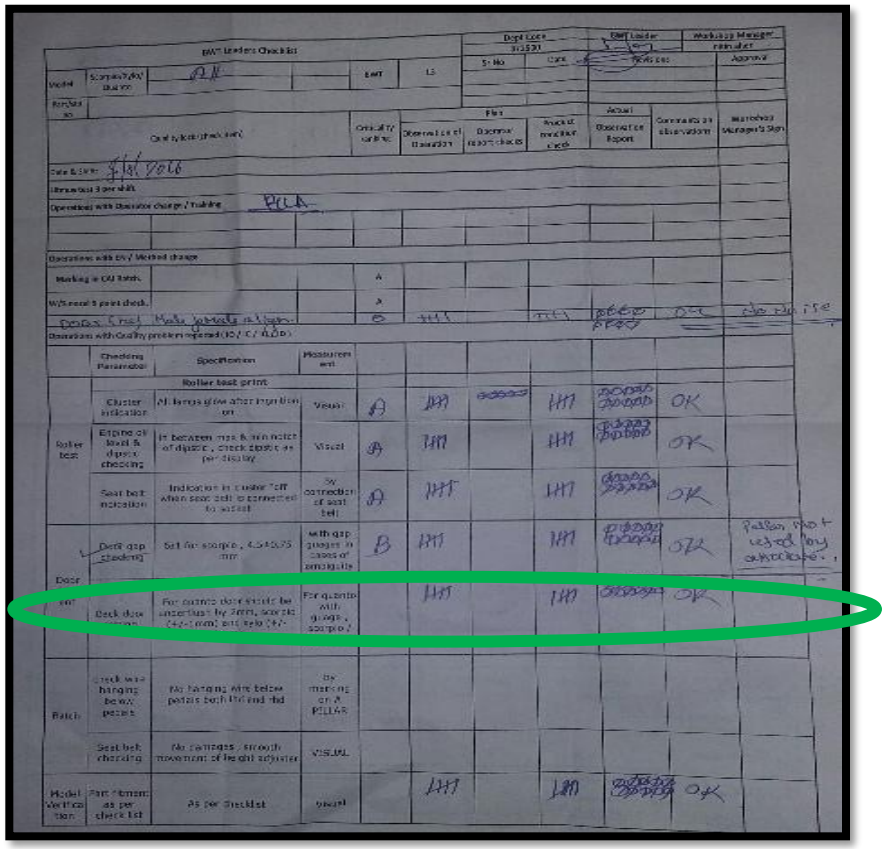
SN	Document	Actual date	Responsibility	Status
1	Fixture history card updated	15.01.19	Nitin/Pravin	Closed

FIXTURE HISTORY CARD					
ZONE:	Z24-Hemming cell				DEPT : MFG ENGG.-SCORPIO BODY
FIXTURE NO.	Z24-20-BD		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME:	Assy Back door 20				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
18/9/14			Inner panel damage	Hinge reinf cu plate mtg modified.	Scrap
19/10/14			Dent -Back door inside	Provided SS guard	CA291095
9/11/14			Damage inner inside at Bottom	Provided guard	CA3P284
20/4/15			BackD oor inner panel damage	Provided guard	
12/10/2015			Spot missing at back door	Provided interlock with gun	PCRN-2274
8/9/2015			Damage Back door Near Hinge Area.	1)Nylon guard shifted for matching with inner pnl. & 2)One piece spring steel guard made.	CA3P 337
4/10/15			Dent Back Door Inside	For gun angle fixture tilted for perpendicular spotting.	CA 3P 343
6/10/2016			Damage Back Door Inside bottom on curve RH.	PU material fitted on resting myler	CA 3P 344
15/4/17			Paint peel off -Fouling Hinge Back door bottom to outer bottom	Correction done at Back door st.20 fixture by 0.5 mm in Z direction hinge reinforcement	
12/9/2018			Damage at inner panle side	Nylon guard provided.	
10/6/2019			Backdoor Fouling Back door hinge to outer panel	Hinge reinf position corrected by 1mm in +Y direction from bottom	
15/1/2020			Back door Sagging / fouling on Striker panel	Qtr inner lower assly fixture position shifted in -Z direction (downward direction) by 0.8 mm at Qtr inner assy & after action Striker found on door latch of V position In lower side upto 0.8 mm	CAI Concern

8. Standardization

8.3 Additional Audit & check

2	Check point added in Daily audit for sustenance of actions.	13.01.19	Satish Patil	Closed



Check Point added in Daily supervisor check sheet for sustenance

9. Summarize & Plan future actions

9.1 Conclusion:

Concern of Back door Sagging & rattle noise reduced by more than 95% by Compensation of 1 mm added at seal gap & Qtr inner lower assly fixture position shifted in -Z direction (downward direction) by 0.8 mm.

9.2 Horizontal Deployment:

All Scorpio models.

9.3 Lessons Learnt:

Things Gone Wrong :

- Initially we suspect only mfg process was suspected as cause of concern.

Things Gone Right :

- Systemic data analysis gives path for Identifying correct & effective actions.

Thank you

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